

Chapter 38 - Graphs

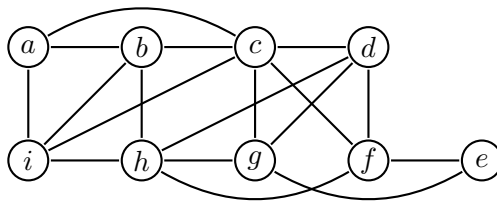
1 pts

1. Derive a formula for the number of edges in K_n , the complete graph with n vertices.

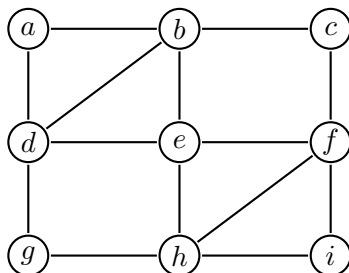
1 pts

2. For each graph below, find an Eulerian circuit, or prove that none exists, and find a Hamiltonian circuit or prove that none exists.

(a)



(b)



1 pts

3. For each of the items below, draw an example of a graph that satisfies the given properties:

(a) A graph with exactly 8 vertices and exactly 3 bridges.

(b) A graph with exactly 7 vertices and no bridges.

(c) A graph with at least 10 vertices, at least 9 edges, and exactly 3 components.

1 pts

4. Draw an example of a graph with up to 4 vertices that is not simple but is regular.

1 pts

5. Draw a graph that has exactly 10 vertices and has an Euler Path. Then, use Fleury's algorithm to find an Euler path in the graph you have drawn.

1 pts

6. Draw a graph that has exactly 8 vertices and has an Euler Circuit.

1 pts

7. Draw any graph which has a Hamilton Circuit but not a Hamilton Path. However, if this is impossible, clearly and succinctly explain why.

1 pts

8. Draw a connected, simple graph with six vertices which has no Euler Circuit, no Euler Path, no Hamilton Circuit, and no Hamilton Path. However, if this is impossible, clearly and succinctly explain why.

- 1 pts 9. Prove that K_n has $n!$ Hamilton paths and $(n - 1)!$ Hamilton circuits.

- 1 pts 10. What is the Hamilton circuit and total weight resulting from the application of the Repetitive Nearest Neighbor Algorithm to the following graph?

